

Linux OS and Scripting

LAB Assignment 2

Name: Vivek B S

Roll No: 261100690034

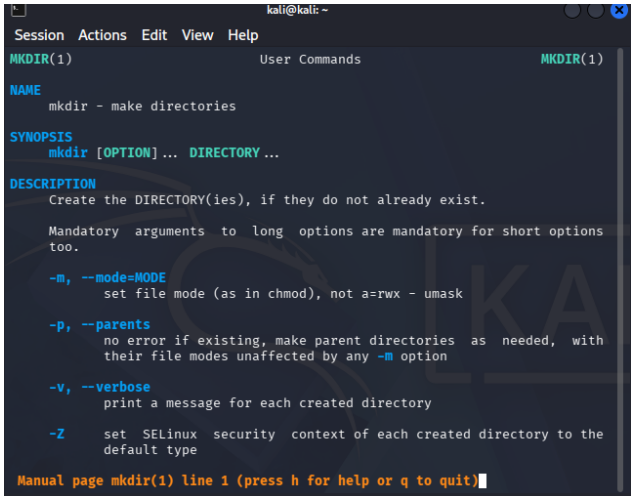
Subject: Linux OS and Scripting

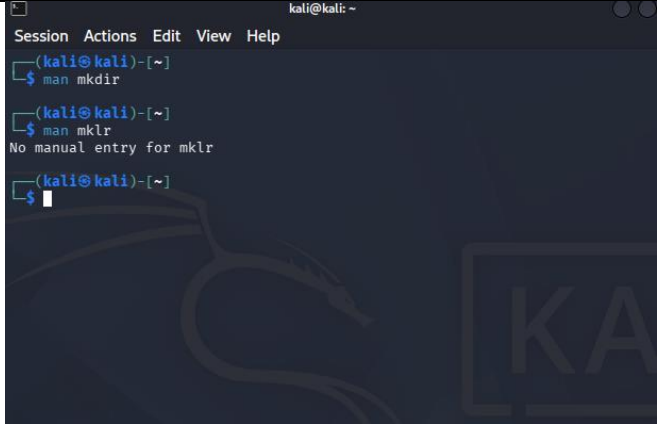
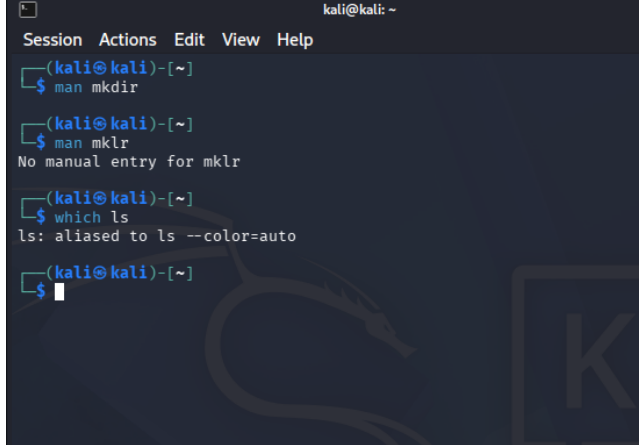
Topic Name:

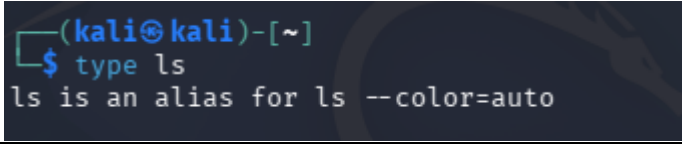
The main aim of this lab session is to provide hands-on experience on

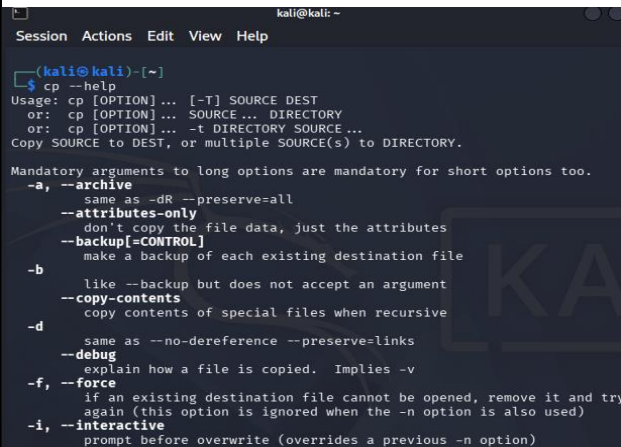
- Getting Help
- Basic Commands
- Navigation
- File System
- Simple shell script
- Globbing
- Extended Globbing
- Filters -I (head,tail,wc)

1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man command	man mkdir	 A screenshot of a terminal window showing the manual page for the 'mkdir' command. The window title is 'kali@kali: ~'. The terminal displays the following content: 'Session Actions Edit View Help', 'MKDIR(1) User Commands MKDIR(1)', 'NAME mkdir - make directories', 'SYNOPSIS mkdir [OPTION]... DIRECTORY...', 'DESCRIPTION Create the DIRECTORY(ies), if they do not already exist. Mandatory arguments to long options are mandatory for short options too. -m, --mode=MODE set file mode (as in chmod), not a=rwx - umask -p, --parents no error if existing, make parent directories as needed, with their file modes unaffected by any -m option -v, --verbose print a message for each created directory -Z set SELinux security context of each created directory to the default type', and 'Manual page mkdir(1) line 1 (press h for help or q to quit)'.

To get manual page for the unknown command	man	man command	man mklr	 <pre> kali@kali: ~ Session Actions Edit View Help (kali@kali)-[~] \$ man mkdir (kali@kali)-[~] \$ man mklr No manual entry for mklr (kali@kali)-[~] \$ </pre>
To know the source file binary	which	which command	which ls	 <pre> kali@kali: ~ Session Actions Edit View Help (kali@kali)-[~] \$ man mkdir (kali@kali)-[~] \$ man mklr No manual entry for mklr (kali@kali)-[~] \$ which ls ls: aliased to ls --color=auto (kali@kali)-[~] \$ </pre>
To know the path of the command	whereis	whereis command	whereis bash	 <pre> (kali@kali)-[~] \$ whereis bash bash: /usr/bin/bash /usr/share/man/man1/bash.1.gz (kali@kali)-[~] \$ </pre>

To know the command is external or internal	Type	type command	type ls	
To get help for the internal command	help	run help - command	run -help cd	<pre>cd [-qslP] [arg] cd [-qslP] old new cd [-qslP] {+ -}n</pre> Change the current directory. In the first form, change the current directory to arg, or to the value of \$HOME if arg is not specified. If arg is '-', change to the previous directory. Otherwise, if arg begins with a slash, attempt to change to the directory given by arg. If arg does not begin with a slash, the behaviour depends on whether the current directory '.' occurs in the list of directories contained in the shell parameter cdpattern. If it does not, first attempt to change to the directory arg under the current directory, and if that fails but cdpattern is set and contains at least one element attempt to change to the directory arg under each component of cdpattern in turn until successful. If '.' occurs in cdpattern, then cdpattern is searched strictly in order so that '.' is only tried at the appropriate point. The order of testing cdpattern is modified if the option POSIX_CD is set, as described in the documentation for the option. If no directory is found, the option CDABLE_VARS is set, and a parameter named arg exists whose value begins with a slash, treat its value as the directory. In that case, the parameter is added to the named directory hash table.
To list out bash commands	compgen	bash Compgen -b	bash Compgen -b	

				<pre> (kali@kali)-[~] \$ bash (kali@kali)-[~] \$ compgen -b : : [alias bg bind break builtin caller cd command compgen complete comptop continue declare dirs disown echo enable eval exec </pre>
To know the usage of the command	--help	Command --help	cp --help	 <pre> kali@kali: ~ Session Actions Edit View Help (kali@kali)-[~] \$ cp --help Usage: cp [OPTION]... [-T] SOURCE DEST or: cp [OPTION]... SOURCE... DIRECTORY or: cp [OPTION]... -t DIRECTORY SOURCE... Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY. Mandatory arguments to long options are mandatory for short options too. -a, --archive same as -dR --preserve=all --attributes-only don't copy the file data, just the attributes --backup[=CONTROL] make a backup of each existing destination file -b like --backup but does not accept an argument --copy-contents copy contents of special files when recursive -d same as --no-dereference --preserve=links --debug explain how a file is copied. Implies -v -f, --force if an existing destination file cannot be opened, remove it and try again (this option is ignored when the -n option is also used) -i, --interactive prompt before overwrite (overrides a previous -n option) </pre>

2. Basic Commands

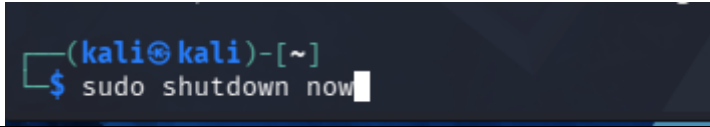
Task	Command Name	Syntax	Example	Screenshots
------	--------------	--------	---------	-------------

To know today's date	date	date	date	<pre>(kali㉿kali)-[~] \$ date Sat Aug 15 08:47:21 AM EDT 2026</pre>
To print calendar	cal	Cal	cal	<pre>(kali㉿kali)-[~] \$ cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 (kali㉿kali)-[~] \$ █</pre>
To print kernel version	uname	uname -r	uname -r	<pre>(kali㉿kali)-[~] \$ uname -r 7.0.12+kali-amd64 (kali㉿kali)-[~] \$ █</pre>
To print default shell	echo	echo \$SHELL	echo \$SHELL	<pre>(kali㉿kali)-[~] \$ echo \$SHELL /usr/bin/zsh</pre>
To print currently logged in user	whoami	whoami	whoami	<pre>(kali㉿kali)-[~] \$ whoami kali</pre>

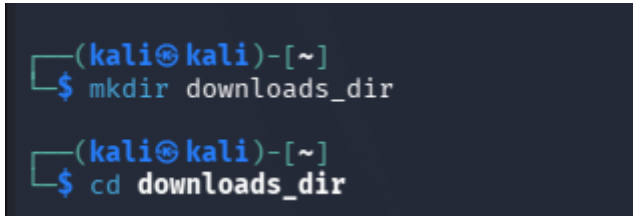
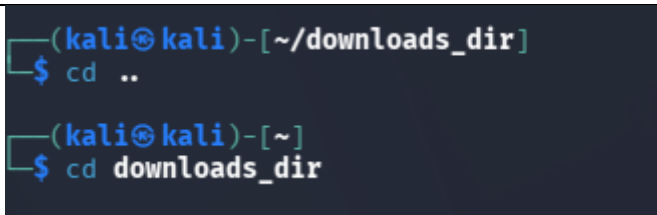
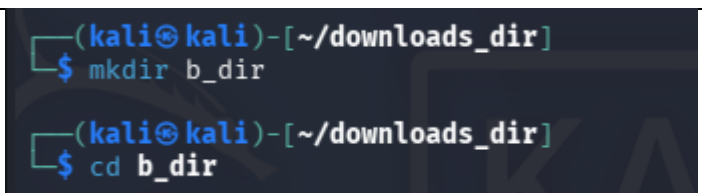
To create shortcut for command	alias	alias name='command'	alias list='ls -i'	<pre> (kali㉿kali)-[~] \$ alias list='ls -i' (kali㉿kali)-[~] \$ list 700776 2 700770 Lab 700669 Pictures 700735 day1 700771 lab1 700671 Projects 700769 Day1 700785 lab1.txt 700666 Public 700663 Desktop 700786 lab4 700780 shrutha 700667 Documents 700777 LinuxOS_2026 700665 Templates 700664 Downloads 700733 Los 700670 Videos 700779 example 700668 Music 700778 lab 1055000 phishing_simulator </pre>
To delete shortcut	unalias	unalias name	unalias list	<pre> (kali㉿kali)-[~] \$ unalias list (kali㉿kali)-[~] \$ list Command 'list' not found, did you mean: command 'mlist' from deb mblaze command 'flist' from deb nmh command 'lift' from deb lift command 'last' from deb wtmpdb command 'hist' from deb loki command 'gist' from deb yorick command 'dist' from deb nmh Try: sudo apt install <deb name> </pre>
To change the timestamp of the file	touch	touch file	touch a.txt ls -l a.txt	<pre> (kali㉿kali)-[~] \$ touch a.txt (kali㉿kali)-[~] \$ </pre>
To clear the screen	clear	Clear	clear	<pre> (kali㉿kali)-[~] \$ clear </pre>

To create empty files	touch	touch filename	touch b.txt	<pre>(kali@kali)-[~] \$ touch b.txt (kali@kali)-[~] \$</pre>
To know disk usage	du	du -sh	du -sh	<pre>(kali@kali)-[~] \$ du -sh 93M . (kali@kali)-[~] \$</pre>
To know free space in the system	df	df -h	df -h	<pre>(kali@kali)-[~] \$ df -h Filesystem Size Used Avail Use% Mounted on udev 888M 0 888M 0% /dev tmpfs 197M 984K 196M 1% /run /dev/sda1 79G 17G 58G 23% / tmpfs 985M 4.0K 985M 1% /dev/shm none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service tmpfs 985M 3.9M 981M 1% /tmp none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service tmpfs 197M 116K 197M 1% /run/user/1000</pre>
To know about the Linux release	cat	cat /etc/os-release	cat /etc/os-release	<pre>(kali@kali)-[~] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.3" VERSION="2026.3" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"</pre>

To show login history	last	last	last	<pre>(kali㉿kali)-[~] \$ last kali tty7 :0 Sat Aug 15 08:25 - still logged in lightdm tty7 :0 Sat Aug 15 08:25 - 08:25 (00:00) kali tty7 :0 Sat Aug 15 05:19 - still logged in lightdm tty7 :0 Sat Aug 15 05:17 - 05:19 (00:01) kali tty7 :0 Tue Aug 11 06:59 - still logged in lightdm tty7 :0 Tue Aug 11 06:58 - 06:59 (00:00) kali tty7 :0 Tue Aug 11 04:52 - still logged in lightdm tty7 :0 Tue Aug 11 04:52 - 04:52 (00:00) postgres pts/2 :0 Mon Aug 10 14:47 - 14:47 (00:00) kali :0 Mon Aug 10 14:46 - still logged in kali tty7 :0 Mon Aug 10 11:29 - still logged in lightdm tty7 :0 Mon Aug 10 11:29 - 11:29 (00:00) kali tty7 :0 Sat Aug 8 11:43 - still logged in lightdm tty7 :0 Sat Aug 8 11:43 - 11:43 (00:00) postgres :0 Tue Jun 16 10:29 - 10:29 (00:00) wtmptdb begins Tue Jun 16 10:29:31 2026</pre>
To check the current user privileges	id	id	id	<pre>(kali㉿kali)-[~] \$ id uid=1000(kali) gid=1000(kali) groups=1000(kali),4(adm),20(dialout),24(cdrom), 25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(ne tdev),102(scanner),118(wireshark),119(kaboxer),968(vboxsf),982(bluetooth),999 (lpadmin)</pre>
To find about the Linux release	lsb_release	lsb_release -a	lsb_release -a	<pre>(kali㉿kali)-[~] \$ lsb_release -a No LSB modules are available. Distributor ID: Kali Description: Kali GNU/Linux Rolling Release: 2026.3 Codename: kali-rolling</pre>
To check system uptime	uptime	uptime	Uptime	<pre>(kali㉿kali)-[~] \$ uptime 09:02:37 up 37 min, 1 user, load average: 0.52, 0.55, 0.56</pre>

To shutdown	shutdown	Sudo shutdown now	Sudo shutdown now	
To restart	reboot	Sudo reboot	Sudo reboot	

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd	cd	
To navigate to the parent directory	cd ..	cd ..	
To navigate to the child directory	cd	cd foldername	

Alternate command to cd	pushd	pushd directory	<pre>(kali@kali)-[~/downloads_dir] \$ pushd b_dir ~/downloads_dir/b_dir ~/downloads_dir (kali@kali)-[~/downloads_dir/b_dir] \$</pre>
To go back to the previous directory	cd -	cd -	<pre>(kali@kali)-[~/downloads_dir/b_dir] \$ cd - ~/downloads_dir</pre>
To go to the root directory	cd /	cd /	<pre>(kali@kali)-[~/downloads_dir] \$ cd / (kali@kali)-[/] \$</pre>

4. File System

Task	Syntax	Command	Screenshot
How to identify the file system	df -T	df -T	<pre>(kali@kali)-[/] \$ df -T Filesystem Type 1K-blocks Used Available Use% Mounted on udev devtmpfs 908640 0 908640 0% /dev tmpfs tmpfs 201640 984 200656 1% /run /dev/sda1 ext4 82083148 17283264 60584336 23% / tmpfs tmpfs 1008184 4 1008180 1% /dev/shm none tmpfs 1024 0 1024 0% /run/credentials/sy stemd-journald.service tmpfs tmpfs 1008184 3936 1004248 1% /tmp none tmpfs 1024 0 1024 0% /run/credentials/ge tty@tty1.service tmpfs tmpfs 201636 116 201520 1% /run/user/1000</pre>

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use **globbing patterns wherever possible**, rather than specifying filenames individually.

- a. List all files ending in .html.

```
ls *.html
```

- b. List all files **other than .html files** using extended globbing.

```
ls !(*.html)
```

- c. List filenames containing at least one numeric digit.

```
ls *[0-9]*
```

- d. List files whose names begin with user followed by a single digit.

```
ls user?.*
```

- e. List files ending in either .txt or .log using a single **extended glob pattern**.

```
ls *.@(txt|log)
```

- f. List files that do **not** end in .txt or .log.

```
ls !(*.txt|*.log)
```

- g. List files beginning with either access or error using extended globbing.

```
ls @(access|error).log
```

- h. List .html files whose filenames contain a numeric digit.

```
ls *[0-9]*.html
```

```

(kali@kali)-[~]
$ bash
(kali@kali)-[~]
$ shopt -s extglob

(kali@kali)-[~]
$ mkdir globbing_practice

(kali@kali)-[~]
$ cd globbing_practice

(kali@kali)-[~/globbing_practice]
$ touch index.html login.html admin.htm report.txt access.log error.log use
r1.txt user2.log file123 backup1.html backup2.txt

(kali@kali)-[~/globbing_practice]
$ ls
access.log  backup1.html  error.log  index.html  report.txt  user2.log
admin.htm  backup2.txt  file123    login.html  user1.txt

(kali@kali)-[~/globbing_practice]
$ ls *.html
backup1.html  index.html  login.html

(kali@kali)-[~/globbing_practice]
$ ls !(*.html)
access.log  backup2.txt  file123  user1.txt
admin.htm  error.log   report.txt  user2.log

```

```

(kali@kali)-[~/globbing_practice]
$ ls *[0-9]*
backup1.html  backup2.txt  file123  user1.txt  user2.log

(kali@kali)-[~/globbing_practice]
$ ls user?.*
user1.txt  user2.log

(kali@kali)-[~/globbing_practice]
$ ls *.@(txt|log)
access.log  backup2.txt  error.log  report.txt  user1.txt  user2.log

(kali@kali)-[~/globbing_practice]
$ ls !(*.txt|*.log)
admin.htm  backup1.html  file123  index.html  login.html

(kali@kali)-[~/globbing_practice]
$ ls @(access|error).log
access.log  error.log

(kali@kali)-[~/globbing_practice]
$ ls *[0-9]*.html
backup1.html

(kali@kali)-[~/globbing_practice]
$

```

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

```
ls @(access|error|auth).log
```

Select all files except notes.txt.

```
ls !(notes.txt)
```

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

```
ls auth.log.@(1|2)
```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
ls malware*[0-9]*.log
```

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed_login or successful_login.

```
ls @(failed_login|successful_login).log
```

Construct a pattern that lists all .log files **except** firewall.log.

```
ls !("firewall.log")
```

```
(kali@kali)~$ mkdir extended
(kali@kali)~$ cd extended
(kali@kali)~/extended$ touch access.log error.log auth.log auth.log.1 auth.log.2 system.log notes.txt
(kali@kali)~/extended$ ls
access.log  auth.log.1  error.log   system.log
auth.log    auth.log.2  notes.txt
(kali@kali)~/extended$ ls @(access|error|auth).log
access.log  auth.log  error.log
(kali@kali)~/extended$ ls !(notex.txt)
access.log  auth.log.1  error.log   system.log
auth.log    auth.log.2  notes.txt
(kali@kali)~/extended$ ls auth.log.@(1|2)
auth.log.1  auth.log.2
```

```
(kali@kali)~/extended$ touch malware1.log malware2.log malwareA.log malware_report.txt normal.log
(kali@kali)~/extended$ ls malware*[0-9]*.log
malware1.log  malware2.log
(kali@kali)~/extended$ touch failed_login.log successful_login.log login_backup.log login.txt firewall.log
(kali@kali)~/extended$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log
(kali@kali)~/extended$ ls !("firewall.log")
access.log  error.log  malware1.log  normal.log
auth.log    failed_login.log  malware2.log  notes.txt
auth.log.1  login_backup.log  malwareA.log  successful_login.log
auth.log.2  login.txt        malware_report.txt  system.log
```

7. Filters

- a. Display the first **10 entries** of login.log.
head -10 login.log
- b. Display the first **5 entries** of access.log.
head -5 access.log
- c. Display the first **15 security events** from security.log.
head -15 security.log
- d. Display the first **20 events** recorded in incident.log.
head -20 incident.log
- e. Display only the **first entry** in login.log.
head -1 login.log
- f. Display the first **25 web requests** recorded in access.log.
head -25 access.log
- g. Display the last **10 entries** of login.log.
tail -10 login.log
- h. Display the last **5 entries** of access.log.
tail -5 access.log
- i. Display the **20 most recent security events** from security.log.
tail -20 security.log
- j. Display the **25 most recent events** from incident.log.
tail -25 incident.log

k. Display only the **last entry** in security.log.

```
tail -1 security.log
```

l. Display the last **50 events** recorded in incident.log.

```
tail -50 incident.log
```

m. Display the contents of login.log starting from **line 41 until the end of the file**.

```
tail -n +41 login.log
```

n. Display the contents of security.log starting from **line 101 until the end of the file**.

```
tail -n +101 security.log
```

o. Find the total number of lines in login.log.

```
wc -l login.log
```

p. Find the total number of lines in access.log.

```
wc -l access.log
```

q. Find the total number of security events in security.log.

```
wc -l security.log
```

r. Determine the total number of events recorded in incident.log.

```
wc -l incident.log
```

s. Count the number of words in login.log.

```
wc -w login.log
```

t. Count the number of bytes in security.log.

```
wc -c security.log
```

- u. Display the **line, word, and byte counts** of access.log.

```
wc access.log
```

- v. Find the line counts of all four .log files using a single wc command.

```
wc -l *.log
```

- w. Display the first **20 entries** of login.log and, from those entries, display only the **last 5**.

```
head -20 login.log | tail -5
```

- x. Display the last **20 entries** of login.log and, from those entries, display only the **first 5**.

```
tail -20 login.log | head -5
```

- y. Display the first **30 entries** of access.log and then display only the **last 10 entries** from that output.

```
head -30 access.log | tail -10
```

- z. Display the last **30 entries** of access.log and then display only the **first 10 entries**.

```
tail -30 access.log | head -10
```

- aa. Display the first **50 events** from security.log and then display the **last 10 events** from those 50.

```
head -50 security.log | tail -10
```

- bb. Display the last **50 events** from security.log and then display the **first 10 events** from those 50.

```
tail -50 security.log | head -10
```

- cc. Display entries from **line 11 to line 20** of login.log.

```
head -20 login.log | tail -10
```

- dd. Display entries from **line 21 to line 30** of login.log.

head -30 login.log | tail -10

ee. Display entries from **line 11 to line 20** of access.log.

head -20 access.log | tail -10

ff. Display entries from **line 31 to line 40** of access.log.

head -40 access.log | tail -10

gg. Display security events from **line 21 to line 30** of security.log.

head -30 security.log | tail -10

hh. Display security events from **line 51 to line 60** of security.log

head -60 security.log | tail -10.

ii. Display incident events from **line 101 to line 110** of incident.log.

head -110 incident.log | tail -10

jj. Display incident events from **line 201 to line 220** of incident.log.

head -220 incident.log | tail -20

kk. Display incident events from **line 491 to line 500**.

head -500 incident.log | tail -10

ll. Take the first **20 entries** of login.log and count how many lines are displayed.

head -20 login.log | wc -l

mm. Take the last **15 entries** of access.log and count how many lines are displayed.

tail -15 access.log | wc -l

nn. Take the first **50 entries** of security.log and count the number of words in those entries.

```
head -50 security.log | wc -w
```

oo. Take the last **25 entries** of security.log and count the number of lines.

```
tail -25 security.log | wc -l
```

pp. Extract lines **21–30** from login.log and count the number of extracted lines.

```
head -30 login.log | tail -10 | wc -l
```

qq. Extract lines **51–60** from access.log and count the number of extracted lines.

```
head -60 access.log | tail -10 | wc -l
```

rr. Extract events **101–150** from incident.log and count how many events are displayed.

```
head -150 incident.log | tail -50 | wc -l
```

ss. Display the first **100 events** from incident.log, take the last **20 events** from that output, and count the number of lines.

```
head -100 incident.log | tail -20 | wc -l
```

```
(kali㉿kali)-[~]
$ mkdir filters
$ cd filters
(kali㉿kali)-[~/filters]
$ for i in {1..100}; do echo "Login Event $i"; done > login.log
(kali㉿kali)-[~/filters]
$ for i in {1..120}; do echo "Access Request $i"; done > access.log
(kali㉿kali)-[~/filters]
$ for i in {1..200}; do echo "Security Event $i"; done > security.log
(kali㉿kali)-[~/filters]
$ for i in {1..500}; do echo "Incident Event $i"; done > incident.log
(kali㉿kali)-[~/filters]
$ ls
access.log  incident.log  login.log  security.log
(kali㉿kali)-[~/filters]
$ head -10 login.log
Login Event 1
Login Event 2
Login Event 3
Login Event 4
```